

EMİNE YILDIRIM

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EDUCATION

Ph.D. in Mathematics,
Université du Québec à Montréal, Canada. August 2015–May 2018
Advisor : Hugh Thomas

MSc. in Mathematics, February 2010–June 2011
Abant İzzet Baysal University, Turkey.
Advisor : Tahire Özen Öztürk

BSc. in Mathematics, September 2006–January 2010
Abant İzzet Baysal University, Turkey.

WORK EXPERIENCE

Assistant Professor May 2026–present
Galatasaray University

R3 Established Researcher August 2025–July 2026
International Center for Mathematical Sciences (ICMS) - Sofia

Research Fellow October 2022–August 2025
University of Leeds, UK

INI-Simons Postdoctoral Fellow October 2021–September 2022
Isaac Newton Institute (INI); University of Cambridge

Coleman Postdoctoral Fellow July 2018–June 2021
Queen's University, Canada

Teaching Assistant January 2014–August 2015
University of New Brunswick, Canada

Lecturer September 2012–December 2013
İstanbul Arel University, Turkey

PAPERS

E. Banaian, W. Kang, E. Kelley, E. Kantarcı Oğuz, E. Yıldırım, *Cluster Expansions from Punctured Orbifolds*, arXiv: 2605.04982.

K. Baur, A. Felikson, D. Prasad, P. Tumarkin and E. Yıldırım, *Growth of infinite frieze patterns of affine type*, arXiv:2603.21157.

- L. Bittmann, P. Jouteur, E. Kantarcı Oğuz, M. Molander, E. Yıldırım, *A mirror deformation of Markov numbers*, arXiv:2602.14802.
- F. Campanini, F. Fedele, E. Yıldırım, *Lattices of pretorsion classes*, arXiv:2511.19223.
- C. Paquette, J. D. Rock, and E. Yıldırım, *Categories of generalized thread quivers*, arXiv:2410.14656.
- K. Baur, L. Bittmann, E. Gunawan, G. Todorov, and E. Yıldırım, *Infinite friezes of affine type D*, Combinatorial Theory, Volume 6, issue 2 (2026), no 12.
- I. Le and E. Yıldırım, *Cluster Categorification of Rank 2 Webs*, arXiv:2402.14501.
- A. Felikson, P. Tumarkin and E. Yıldırım, *Polytopal realizations of non-crystallographic associahedra*, Algebraic Combinatorics, Volume 8, issue 1 (2025), 17-28.
- E. Kantarcı Oğuz and E. Yıldırım, *Cluster Expansions: T-walks, Labeled Posets and Matrix Calculations*, Journal of Algebra (2025), Vol. 669, 183-219.
- R. Marczinzik, H. Thomas, and E. Yıldırım, *On the interaction of the Coxeter transformation and the Rowmotion bijection*, Journal of Combinatorial Algebra 8 (2024), 359–374.
- C. Paquette and E. Yıldırım, *A Completion of Discrete Cluster Categories of type A*, Transactions of the London Mathematical Society (2021), 8(1), 35–64.
- V. Bazier-Matte, N. Chapelier-Laget, G. Douville, K. Mousavand, H. Thomas, and E. Yıldırım, *ABHY associahedra and Newton polytopes of F -polynomials for cluster algebras of simply laced finite type*. Journal of the London Mathematical Society (2) 2024, 109 (1), Paper No. e12817, 27.
- V. Bazier-Matte, G. Douville, A. Garver, R. Patrias, H. Thomas, and E. Yıldırım, *Leading Terms of SL_3 Web Invariants*. International Mathematics Research Notices, (2020), rnaa110.
- T. Brüstle, G. Douville, K. Mousavand, H. Thomas, and E. Yıldırım, *On the combinatorics of gentle algebras*. Canadian Journal of Mathematics (2020), 72 (6), 1551-1580.
- E. Yıldırım, *Coxeter transformation on cominusculé posets*. Algebra and Representation Theory (2019), 22 (3), 699-722.
- T. Özen and E. Yıldırım, *$\mathcal{X}$ -Injective and $\mathcal{X}$ -Projective Complexes*. Bulletin of Iranian Mathematical Society (2016), 42 (5), 1221-1235.
- OTHER**
- E. Yıldırım, *The Math Solidarity Platform*, Eur. Math. Soc. Mag. 139, pp. 13–14, 2026.

E. Yildirim, *Pretorsion Theories*, Oberwolfach Report for Representation Theory of Quivers and Finite-Dimensional Algebras, 8-13 February 2026.

E. Yildirim, *Cluster Expansion via matrices*, Oberwolfach Report for Cluster Algebras and Its Applications, 14-19 January 2024.

E. Yildirim, *Webs and $Gr(3,n)$* , Oberwolfach Report for the Arbeitsgemeinschaft: Cluster Algebras, 8-13 October 2023.

K. Baur and E. Yildirim, *Cluster Algebras*, Encyclopedia of Mathematical Physics 2nd edition, book title: Reference Module in Materials Science and Materials Engineering, 2023.

AWARDS/ GRANTS

- Bulgarian National Science Foundation grant (Contract No: KP-06-N92/5), ICMS - Sofia.
- International Mobility Grant, University of Leeds.
- Academic Development Fellowship, University of Leeds.
- London Mathematical Society Scheme 5: ‘Collaborations with Developing Countries’ grant.
- International Congress of Mathematicians 2022 travel grant (award not used since the conference moved to online).
- Invitation/Fund to Hausdorff Junior Trimester Program on ‘New Trends in Representation Theory’ (award declined due to COVID-19/travel restrictions).
- Queen’s University Post-Doctoral Travel Award, 2018.
- Institut des Sciences Mathématiques (ISM) Scholarships for Outstanding PhD Candidates, 2018.
- ISM Travel Grant, 2018.
- ISM Graduate Scholarships at UQAM from 2016 to 2018.
- Tuition Exemption Scholarships at UQAM from 2015 to 2018.

ORGANIZING

I am/was a co-organizer of the following conference and seminars.

- Quantum Groups and Cluster Algebras II, ICMS - Sofia, August 2026.
- Celebrating Women in Representation Theory, ICMS - Sofia, May 2026.
- Farey’s legacy in frieze patterns and discrete geometry, ICMS - Edinburg, April 2026.
- Quantum Groups and Cluster Algebras, ICMS - Sofia, December 2025.
- Inclusion, Diversity, and Equity in Algebra, University of Leeds, October 2025.
- AESIM Summer School on Markov Numbers and Cluster Algebras, İstanbul, August 2025.
- CLAN (Clusters and their Applications in the North), 2024-2025.
- Algebra seminars, University of Leeds, February 2023 - June 2024.

- The Advances in Representation Theory of Algebras IX, Queen's University, June 2023.
- Cluster algebras and Representation Theory Seminars at Isaac Newton Institute, September-December 2021.
- Virtual Advances in Representation Theory of Algebras 2021 - Online Conference in Honor of Andrzej Skowroński, May 17-May 28 2021.
- Algebra & Geometry Working Seminars, Queen's University, 2020-2021.
- Geometry and Representation Theory Seminars, Queen's University, 2019-2020.
- Quantum Groups Learning Seminars, Queen's University, 2019-2020.

TEACHING

University of Leeds, United Kingdom:

- **(Advanced) Commutative Rings and Algebraic Geometry in 2024 & 2025**

I was a module leader of the course which consists of a mix of undergraduate and master students.

- **Projects in Mathematics in 2022-2023**

This is a year long course for students to carry out independent research. I co-supervised 5 students for different projects in the research area of "*Friezes*."

Queen's University, Kingston, Canada:

- **Vector Calculus** in the Fall semesters of 2018; 2019; 2020.

I was the only instructor, thus the coordinator, of this course each semester.

- **Rings and Fields** in the Winter semester 2019.

This was a coordinated course and I was responsible from preparing and giving lectures for my section.

- **Linear Algebra** for the Winter semester 2020; 2021.

This was a coordinated course and I was the coordinator of the online course in 2021.

University of New Brunswick, Fredericton, Canada:

- From January 2014 to June 2015, I worked as a teaching assistant for introductory calculus classes.
- In addition, I worked as a tutor at the Math Help Center.

İstanbul Arel University, İstanbul, Türkiye:

- From September 2012 to December 2013, I was a full time lecturer.

SUPERVISING

- Brennan Smith, *Polytopes: A Combinatorial Approach*, 2025, Master project at the University of Leeds.
- Alex Southworth, *Rowmotion*, 2025, Summer Bursary Scheme at the University of Leeds.
- Julia Hörmayer, *Cluster Categories from non-uniform Postnikov diagrams*, 2023-2024, PhD supervision at the University of Leeds.
- İlknur Öztürk, *Root Systems and Reflection Groups*, summer 2023, Directed Reading Program Türkiye.
- Temi Abbass, *Triangulations and Friezes in three dimensions*, summer 2022, In2Research Program at the University of Cambridge.
- Zeliha Sevgi, *Quiver Representations*, summer 2022, Directed Reading Program Türkiye.
- Aysel Şahin, *Conway-Coxeter Friezes and related combinatorics*, summer 2021, Directed Reading Program Türkiye.
- Elifnaz Gülsen, *Quiver representations and their interaction with algebra and combinatorics*, one semester in 2019, Directed Reading Program Queen's University.
- Molly Liu, *Cluster algebras and combinatorics*, one semester in 2019, Directed Reading Program Queen's University.

ACTIVITIES

I am/was a

- EDI postdoc representative in the University of Leeds, 2024-2025.
- Jury member of the PhD defense by Tal Gottesman.
- co-leader at the workshop “Women in Noncommutative Algebra and Representation Theory 4”, Banff International Research Station, March 23 – 28, 2025.
- host In2Stem Program for 4 high school students at the University of Leeds between 5-9 August 2024.
- host in the In2Science Program for 6 high school students at the University of Leeds between 7-11 August 2023.
- mentor in Directed Reading Program Türkiye in the summer of 2021, 2022 & 2023.
- co-founder of the Maths Support Platform which provided academic resources, support and mentoring in math for undergraduate students who were affected by the major earthquake of February 2023 in Türkiye.
- mentor in Cambridge for 3 female undergraduate students in 2022.
- volunteer in the Mathematics Festival in Istanbul, 14-15 May 2022.

Research Visits

- Hugh Thomas, Université du Québec à Montréal, 7-10 April 2026.
- Charles Paquette and Ivan Dimitrov, Queen's University, 30 March-6 April 2026.
- Eleonore Faber, University of Graz, 25-30 January 2026.
- Daisie Rock, University of Ghent, 17-24 May 2025.
- Léa Bittmann, University of Strasbourg, 30 September - 4 October 2022.
- Charles Paquette, Queen's University, 26 May - 26 June 2023.
- Esther Banaian, Aarhus University, 14-18 February 2023.
- Léa Bittmann, University of Edinburgh, 29 November - 1 December 2022.
- Anna Felikson and Pavel Tumarkin, Durham University, 15-18 July 2022.
- Alastair King, University of Bath, 3-5 May 2022.
- Ana Garcia Elsener, University of Glasgow, 9-12 April 2021.
- Ibrahim Assem, University of Sherbrooke, 30 June - 6 July 2019.
- Ralf Schiffler, University of Connecticut, 29 October - 3 November 2017.

PRESENTATIONS

- *Thread quivers*, Computable Invariants of quiver representations, Lorentz Center, Leiden, July 2026.
- *Thread quivers and their representation categories*, International Conference on Representations of Algebras (ICRA) 2026, Grenoble, June 2026.
- *Wonderful Wizard of Markov numbers*, Sofia One-Board Seminar, IMI-BAS, May 2026
- *Pretorsion Classes*, Celebrating Women in Representation Theory, IMI-BAS, May 2026.
- *A mirror deformation of Markov numbers*, Farey's legacy in frieze patterns and discrete geometry, ICMS Edinburgh, April 2026.
- *A mirror deformation of Markov numbers*, LaCIM Seminar, Université du Québec à Montréal, April 2026.
- *Lattices from representation theory of algebras*, Algebra and Geometry Seminar, Queen's University, March 2026.
- *Categorification of webs beyond Plücker coordinates*, Cluster Algebras, Webs, and Canonical Bases, Banff International Research Station, March 2026.
- *Pretorsion theories*, Representation Theory of Quivers and Finite-Dimensional Algebras, Oberwolfach, February 2026.
- *Pretorsion theory in module categories*, Algebra & Geometry Seminar, University of Graz, January 2026.
- *Lattice of Pretorsion classes*, Flash Talks in Representation Theory VI, January 2026.
- *Cluster Structures, Lecture Series I, II, III*, Summer School on Markov Numbers and Cluster Algebras, August 2025.

- *Understanding algebras via geometry and combinatorics, Lecture Series I, II, III*, Rings, Modules and Algebras 11th edition, Bursa Technical University, July 2025.
- *Frieze patterns of affine type D* AlCoVE: an Algebraic Combinatorics Virtual Expedition, May 2025.
- *Growth of frieze patterns*, Frieze patterns in algebra, combinatorics and geometry, CIRM, May 2025.
- *Posets in the combinatorics of cluster algebras*, Algebra seminar, University of Georgia, April 2025.
- Categorification of webs, Algebra seminar, University of York, February 2025.
- *Getaways to Friezes*, Poster presentation in the Okinawa Institute of Science and Technology (OIST), November 2024.
- *Surface triangulations and Friezes*, Pure Maths Seminar, University of East Anglia, November 2024.
- *Thread quivers*, Women in Representation Theory: quivers, mutations, and beyond, Bielefeld University, September 2024.
- *From Triangulations to Friezes and Cluster Algebras*, Mathematics Department Seminar at Yeditepe University, May 2024 (online).
- *Thread quivers*, Pure Maths Seminar at Lancaster University, May 2024.
- *Exploring Combinatorial Windmills across Algebraic Landscape*, Algebra and Number Theory seminar at the University College Dublin, April 2024.
- *Matrix computation of cluster expansions*, Continued Fractions and SL_2 -tilings, Durham University, March 2024.
- *Why the Return to Pictures in Algebra?*, Pure maths colloquium, University of Sheffield, March 2024.
- *Cluster Expansion via matrices*, Cluster Algebra and Its Applications, Oberwolfach, January 2024.
- *Webs and $Gr(3, n)$* , Arbeitsgemeinschaft: Cluster Algebras, Oberwolfach, October 2023.
- *Friz Desenlerinin Cebirsel Yorumu*, 35. National Mathematics Symposium, Trakya University, Turkey, September 2023.
- *Friezes*, Combinatorial Representation Theory in Leeds, University of Leeds, July 2023.
- *Growth of frieze patterns*, Cluster Structures in the North 2023, University of Central Lancashire, Preston, June 2023.
- *Cluster Expansions via matrices*, Combinatorial aspects of Representation Theory, Norwegian Academy of Sciences and Letters in Oslo, March 2023.
- *Computing Cluster Expansions via matrices*, Glasgow Algebra and Number Theory seminar, University of Glasgow, March 2023.
- *Cluster Expansion via matrices*, Aarhus Homological Algebra Seminar, Aarhus University, February 2023.

- *Categories from Poset Quivers*, Algebra seminar, University of Manchester, December 2022.
- *Poset Quivers*, Algebra seminar, University of Leeds, October 2022.
- *Poset Quivers*, Geometric and combinatorial methods in homological algebra, Aarhus University, July 2022.
- *Cluster-tilting subcategories*, Bridges between representation theory and algebraic geometry: Singularities, friezes and cluster categories, University of Leeds, May 2022.
- Cluster categories: classical, discrete and beyond, Algebra seminar, University of Bath, May 2022.
- What is a connection between cluster algebras and friezes?; with Emily Gunawan, The Banff International Research Station, April 2022.
- *Periodic actions*, Bilkent Algebra Seminar, February 2022 (online).
- *Periodic actions on distributive lattices and counterparts in algebra*, FD Seminar, November 2021 (online).
- *Cluster structures*, New developments in Representation Theory arising from Cluster Algebras, Isaac Newton Institute, September 2021 (online).
- *The manifestations of Rowmotion in Representation Theory of Algebras*, Summer School of Algebraic Combinatorics, June 2021 (online).
- *Composition of the rowmotion and the Coxeter transformation*, Virtual Advances in Representation Theory of Algebras, May 2021 (online).
- *Webs and Modules*, TRAC Online Seminars, University of Sherbrooke, April 2021 (online).
- *Generalized Associahedra*, Algebra Seminar, University of Leeds, March 2021 (online).
- *Tilting theory and Quiver mutation*, LPA Seminars, Turkey, February 2021 (online).
- *Discrete cluster categories of type A and beyond*, Flash Talks in Representation Theory, NTNU, January 2021 (online).
- *The orbits of the Coxeter Transformation and Rowmotion for cominuscul posets*, Dynamical Algebraic Combinatorics, Banff International Research Station, October 2020 (online).
- *Graphs and Combinatorics in Representation Theory of Algebras.*, Queen's University Colloquium, October 2020 (online).
- *Associahedra and Quiver Representations*, Gebze Technical University, October 2020 (online).
- *The Coxeter Transformation and Rowmotion for cominuscul posets*, Kentucky Discrete Math Seminar, September 2020 (online).
- *Cluster categories*, Uluslararası Cebirsel Geometri Neşesi, 2020 (online).
- *Cluster categories capturing combinatorics of infinite cluster variables*, Algebra Seminar, University of Sherbrooke, Quebec, February 2020.

- *The Coxeter Transformation and Promotion*, Algebra Seminar, University of Toronto, Toronto, January 2020.
- *A completion of discrete cluster categories of type A*, Seventh Conference on Geometric Methods in Representation Theory, University of Missouri, Columbia, November 2019.
- *Rowmotion and Coxeter transformation*, Applied Algebra Seminar, York University, Toronto, September 2019.
- *A slight extension of discrete cluster categories*, Cluster Algebras and Quiver Representations, Centre de Recherches Mathématiques (CRM), August 2019.
- *Cluster categories for an infinite rank cluster algebra*, Summer Day of Representation Theory of Algebras, Université de Sherbrooke-Bishop's University, QC, July 2019.
- *The bounded derived category for cominuscule posets*, Maurice Auslander Distinguished Lectures and International Conference, Woods Hole, MA, April 2019.
- *The Web Basis and Preprojective Algebras*, American Mathematical Society Sectional Meeting, Hartford, CT, April 2019.
- *Associahedra and Newton Polytopes of F-Polynomials*, Cluster structures in geometry, physics, combinatorics and representation theory, Jerusalem, December 2018.
- *The bounded derived category for cominuscule posets*, Algebra seminar, Queen's University, Kingston, ON, September 2018.
- *Generalized Associahedra and Newton polytopes of F-polynomials*, Algebra seminar, Carleton University, Ottawa, ON, September 2018.
- *Associahedra via quiver representations*, Canadian Mathematical Society Summer Meeting, Fredericton, NB, June 2018.
- *The periodicity of the Coxeter transformation*, The XXI^e Colloque panquébécois des étudiants de l'Institut des Sciences mathématiques (ISM), Université de Sherbrooke, QC, May 2018.
- *The Coxeter transformation on cominuscule posets*, Séminaire de combinatoire, Université du Québec à Montréal, LaCIM, QC, February 2018.
- *The Coxeter transformation on cominuscule posets*, Department seminar, Mimar Sinan University, Istanbul, December 2017.
- *The Coxeter transformation on grid posets*, Department seminar, Istanbul Technical University, Istanbul, December 2017.
- *Cluster Monomials in Web Basis*, Algebra seminar, University of Sherbrooke, QC, November 2017.
- *Cluster Monomials in Web Basis*, Cluster algebras seminar, University of Connecticut, CT, November 2017.
- *Coxeter transformation on cominuscule posets*, Cluster algebras seminar, University of Connecticut, CT, October 2017.
- *Coxeter transformation on cominuscule posets*, Combinatoric seminar, University of Minnesota, MN, September 2017.

- *Auslander-Reiten translation on cominuscule posets*, Advances in Representation Theory of Algebras VI: Geometry and Homology (ARTA), Luminy, September 2017.
- *Periodic behavior of Auslander-Reiten translation*, The Mathematical Congress of the Americas (MCA), Montreal, QC, July 2017.
- *On the combinatorics of gentle algebras*, XXVIIIth Meeting on Representation Theory of Algebras, Sherbrooke, QC, September 2016.
- *Coxeter transformation of the poset of order ideals in a grid*, International Conference on Representations of Algebras, Syracuse, NY August 2016.
- *Cluster Algebras*, Graduate Seminar, University of New Brunswick, NB, April 2015.
- *Two dimensional Yetter-Drinfeld modules over $U_q(sl_2)$ are trivial*, Hopf Algebras in (Co)Action workshop, NB, June 2014.
- *Monomial ideals of path algebras*, Research Seminars, University of New Brunswick, NB, March 2014.

**SELECTED
RECENT
CONFERENCES &
WORKSHOPS**

- Computable Invariants of quiver representations, Lorentz Center, Leiden, July 2026.
- International Conference on Representations of Algebras (ICRA), Grenoble, France, June-July 2026.
- Farey's legacy in frieze patterns and discrete geometry, ICMS Edinburgh, April 2026.
- Cluster Algebras, Webs, and Canonical Bases, Banff International Research Station, March 2026.
- Representation Theory of Quivers and Finite-Dimensional Algebras, Oberwolfach, February 2026.
- Flash Talks in Representation Theory VI, January 9, 2026.
- Beyond Permutahedra and Associahedra, CIRM, December 2025.
- Frieze patterns in algebra, combinatorics and geometry, CIRM, May 2025.
- Women in Noncommutative Algebra and Representation Theory 4 (WINART4), BIRS, Calgary, March 2025.
- Perspective on Markoff Numbers, BIRS, Calgary, January 2025.
- Lattice Theory, Banff International Research Station (BIRS), Calgary, January 2025.
- New developments in representation theory of algebras: Algebraic, combinatorial and geometric aspects, Okinawa Institute of Science and Technology (OIST), November 2024.
- Clusters and their Applications in the North (CLAN), University of Central Lancashire, Preston in June 2024 & University of Glasgow in February 2024.
- Current Trends in Representation Theory, Cluster Algebras and Geometry, Centre International de Rencontres Mathématiques (CIRM), Luminy, November 2023.

- Arbeitsgemeinschaft: Cluster Algebras, Oberwolfach, October 2023.
- Functor Categories, Model Theory, and Constructive Category Theory, University of Málaga, July 2023.
- Advances in Representation Theory of Algebras IX, Queen's University, Kingston, June 2023.
- ARTIN meeting, the University of Manchester, March 2022.
- Representation Theory and Geometry, Queen's University - RMC, February 2022 (online).
- Combinatorial Algebra meets Algebraic Combinatorics, 19th Annual Meeting, The Fields Institute, January 2022 (online).
- New developments in representation theory arising from cluster algebras, Isaac Newton Institute, September 2021 (online).
- Flash Talks in Representation Theory, NTNU, January 2021 & 2023 (online), 2025.
- Meeting on Representation Theory of Algebras – Corona Edition, Sherbrooke, QC, September 2020 (online).
- Dimers in Combinatorics and Cluster Algebras, University of Michigan, August, 2020 (online).

SKILLS

Languages: Turkish (native), English (fluent), French (intermediate).

Software: SAGE, GAP, PYTHON.

Art: Painting, Photography.